6.4.3 Radioactivity

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	radioactive decay; spontaneous and random nature of decay	*M1.3*
(b)	**(i)** α-particles, β-particles and γ-rays; nature, penetration and range of these radiations	
	(ii) techniques and procedures used to investigate the absorption of α-particles, β-particles and γ-rays by appropriate materials	**PAG7**
(c)	nuclear decay equations for alpha, beta-minus and beta-plus decays; balancing nuclear transformation equations	
(d)	activity of a source; decay constant λ of an isotope; $A = \lambda N$	**THIS WILL ALSO REQUIRE KNOWLEDGE OF 5.1.4(a)**
(e)	**(i)** half-life of an isotope; $\lambda t_{1/2} = \ln(2)$	
	(ii) techniques and procedures used to determine the half-life of an isotope such as protactinium.	**PAG7**
(f)	**(i)** the equations $A = A_0 e^{-\lambda t}$ and $N = N_0 e^{-\lambda t}$, where A is the activity and N is the number of undecayed nuclei	*M3.12*
	(ii) simulation of radioactive decay using dice	
(g)	graphical methods and spreadsheet modelling of the equation $\frac{\Delta N}{\Delta t} = -\lambda N$ for radioactive decay	HSW3 Using spreadsheets to model the radioactive decay of nuclei. *M0.5, M2.5, M3.9*
(h)	radioactive dating, e.g. carbon-dating.	